

# JUSTIN KIM

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## Education

**University of Wisconsin, Madison | GPA: 3.76**

**Expected Graduation: Dec 2026**

*Bachelor of Science in Computer Science & Data Science*

*Madison, WI*

- **Relevant Coursework:** Algorithms, Artificial Intelligence, Big Data Systems, Database Management Systems, Computer Networks, Machine Organization, Data Science Modeling I & II, Linear Algebra, Discrete Mathematics

## Work Experience

**Capital One**

**June 2026 – Aug 2026**

*Software Engineer Intern*

*Chicago, IL*

- Built an AI-assisted authoring system that enables non-engineers to generate credit card fulfillment workflow configs from plain-English requirements, reducing manual config authoring from 2+ days to under 1 hour.
- Designed a GraphRAG pipeline that retrieves relevant workflow actions, grounds LLM generation on approved action schemas, and validates outputs with state-reference checks before configs are loaded.
- Shipped a Go/AWS Lambda streaming API with TypeScript CDK infrastructure, adding observability, private networking, and TLS while cutting infrastructure code footprint by 55% versus the reference implementation.

**MadLease**

**Jan 2024 – Present**

*Co-Founder & Founding Engineer*

*Madison, WI*

- Founded and deployed a UW–Madison sublease marketplace with 200+ accounts, 90+ listings, and 5+ building partners.
- Boosted listing visibility across 25+ crawler types by serving structured metadata via Vercel Edge Middleware, while routing real users directly to the React SPA.
- Eliminated backend image processing for 1,000+ uploads by routing browser-direct S3 transfers through presigned URLs, CloudFront delivery, and Firebase UID/IP rate limiting.

**Integral**

**May 2025 – Aug 2025**

*Full Stack Software Engineer Intern*

*Madison, WI*

- Led development of a production-ready crash photo capture and insurance intake platform, creating a company-wide reusable OCR module that extracted VIN, license plate, and odometer fields with 90%+ accuracy.
- Eliminated polling for estimate negotiation data by building a Pub/Sub database listener that propagated updates across multiple internal services in real time.

**Denari**

**Feb 2025 – May 2025**

*Software Engineer, Client Project*

*Madison, WI*

- Built a citation-grounded RAG pipeline for tax document search, improving answer accuracy by 8% over a GPT-4o-mini baseline across 2,200+ CPA exam benchmark questions using OpenAI embeddings, TimescaleDB/pgvector, and AWS S3.

**Astro AI Trading**

**Jan 2024 – Dec 2024**

*Machine Learning Engineer*

*Remote*

- Led a 3-engineer team at an early-stage fintech startup building sentiment-augmented stock forecasting models trained on market data and 60+ daily news/social sources, reducing held-out RMSE by 32% over a price-only baseline.
- Built a BERT + PCA + KMeans article routing pipeline that clustered financial news by sector, improving sentiment signal quality for downstream forecasting models.

## Projects

**QuizClash** | [Devpost](#) | [Source Code](#)

**NodeJS | React | WebSocket | AWS**

- Developed a real-time multiplayer quiz app converting PDFs and YouTube videos into Kahoot-style games with synchronized WebSocket gameplay and public/private rooms.
- Automated question generation with a 5-stage ML pipeline (OCR, Whisper, T5, RoBERTa, Sense2Vec) for content ingestion, question generation, answer extraction, and distractor selection, reaching 75% manually evaluated relevance.

## Technical Skills

**Languages:** Python, Java, JavaScript, TypeScript, C, SQL, HTML5, CSS

**Developer Tools:** Git, Docker, AWS (S3, EC2, IAM), Postman, Jira, WireMock, Pub/Sub

**Libraries/Frameworks:** React, Spring Boot, Node/Express, PostgreSQL, Flask, PyTorch, Angular, Scikit-Learn

**Concepts:** Machine Learning, Full Stack Development, Backend, Frontend, Data Science, Agile Methodologies